

Reproducing VISAtlas

An Image-Based Exploration and Query System for Large Visualization Collections

September 25, 2026

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Context



Venue

- **IEEE Transactions on Visualization and Computer Graphics** (TVCG) — the field's flagship journal
- Presented at **IEEE VIS 2023**, TVCG track (accepted Dec. 2022)
- Ships with an open dataset (17,842 images) and public code

Authors — all at HKUST(GZ)

- **Yilin Ye** — PhD student, first author, built the system
- **Rong Huang** — PhD student, architecture / computational aesthetics background
- **Wei Zeng** — Assistant Professor, corresponding author, group lead

Where are they now? Yilin Ye's PhD thesis frames **VISAtlas as Study 1** of a broader line on embedding-based visual exploration, continued in **ModalChorus** (2024) and **AKRMap/DKMap** (2025–26). Rong Huang co-authored a 2025 follow-up applying the same method to human-vs-AI paintings. Wei Zeng continues to lead the group at HKUST(GZ). *(See the citation-provenance timeline shortly.)*

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Motivation & Research Questions

“The visualization collections are inherently diverse in many perspectives: from different data sources, by different research groups, and in different manners.”

“Existing systems for visualization images ... are mainly based on extrinsic attributes including author names and publication years.”

“More importantly, attribute-based systems neglect the intrinsic property, i.e., visual appearance of visualizations that is also perceived by humans.”

Although visual appearance is directly perceived by humans, it has often been overlooked as an analyzable property of visualization collections.

Y. Ye, R. Huang, W. Zeng. VISAtlas. IEEE TVCG 30(7):3224–3240, 2024.

“Which visualization collection shall be used?

Are there any similar visualizations in the collection?

What visual features are prominent in a visualization?

What visual features are frequently composited in visualizations?”

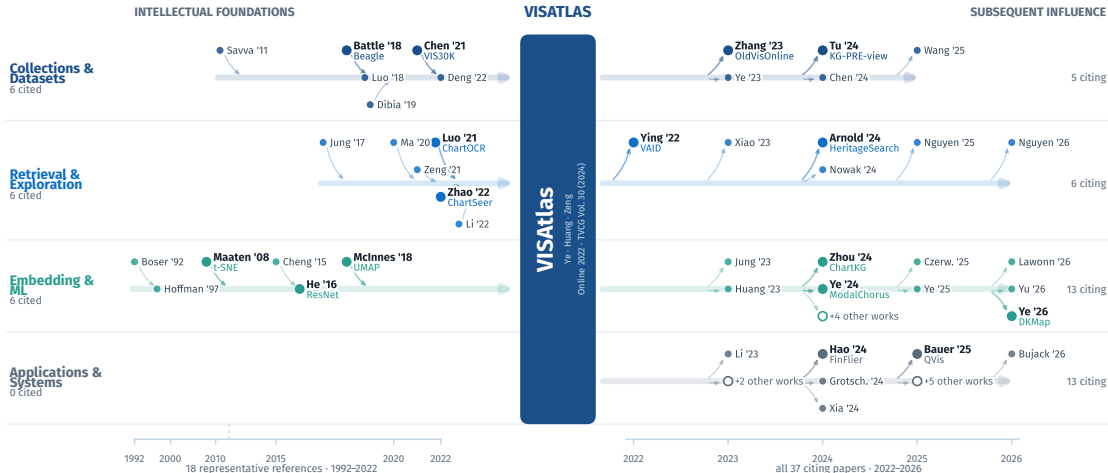
VISAtlas represents the visual appearance of visualizations as interpretable embeddings, enabling users to explore, compare, and query visualization collections based on visual features rather than only extrinsic metadata.

Y. Ye, R. Huang, W. Zeng. VISAtlas. IEEE TVCG 30(7):3224–3240, 2024.

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Related Work & Contributions

Related Work: Citation Provenance



Where did VISATlas come from intellectually? · Which research themes contributed to it? · Where did its influence propagate afterward?

- 1 A **neural image embedding** for chart images: a CNN (ResNet50) trained with a *triplet loss* on a new 17,842-image dataset, turning any chart into an 11-dimensional, human-readable “type vector”
- 2 A **coordinated multiple-view (CMV) interface**: a radial *Embedding Overview*, an *Embedding Histogram*, and a *Visualization Gallery* for query-by-example
- 3 Three **case studies** showing the system can compare collections, surface composite chart designs, and retrieve similar designs from real or hand-drawn input

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System & Method

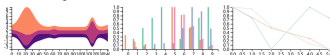


VISAtlas System Overview

What goes in?

Diverse visualization images

Synthetic Visualizations



Diverse chart types, data distributions, and colormaps

Real-World Visualizations

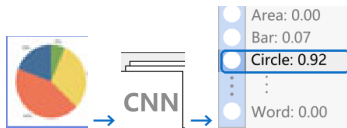


Web-crawled visualizations reflecting realistic usage

Diverse visualizations

What does VISAtlas do?

Encode visual appearance



Human-interpretable semantic embedding

Interpretable semantic representation

Reproducing VISAtlas

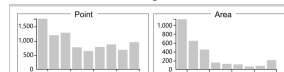
What can users do?

Explore



Collection structure

Compare



Embedding distributions

Query



Similar designs

Visual analytics tasks

Data Collection

Controlled coverage + realistic diversity

Synthetic Visualizations

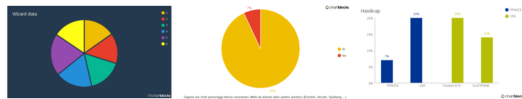
Controlled coverage



Real-World Visualizations

Realistic diversity

Beagle — Web practice



VIS30K — Research practice

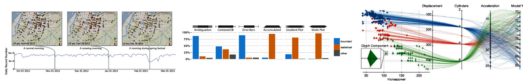


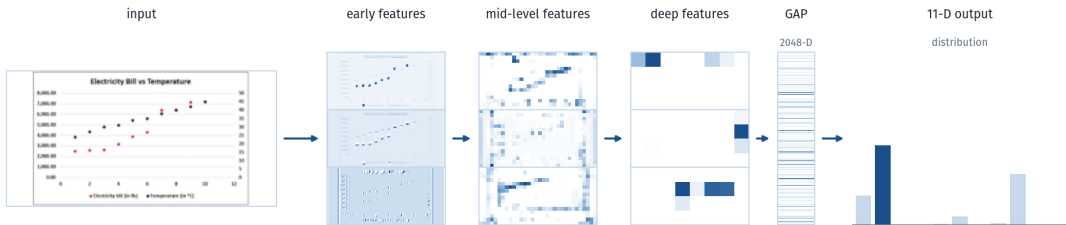
Chart type × Data distribution × Colormap

Synthetic coverage + real-world diversity → a comprehensive visualization collection

Examples from the released VISAtlas resources (github.com/yilinye/VisAtlas-Code)

Image Embedding via CNN

From pixels to feature maps to an 11-D representation



ResNet50 transforms spatial patterns into a compact numerical embedding.

From Triplet Metric Learning to VISAtlas

From identity-based metric learning to visualization-type-aware embeddings

Triplet Metric Learning — FaceNet



$$\|f(A) - f(P)\|_2^2 + \alpha < \|f(A) - f(N)\|_2^2$$

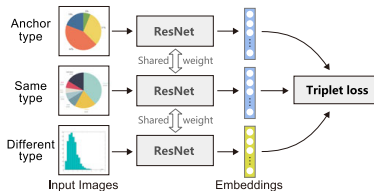
Pull close · Push apart



Triplet learning turns **semantic identity** into **geometric separation**.

Based on FaceNet (Schroff et al., CVPR 2015, Fig. 3); scatter plot is schematic.

VISAtlas: Type Supervision + Triplet Learning



$$L = L_{CE} + \beta L_{TR}$$

$$L_{CE}: q_n \approx p_n$$

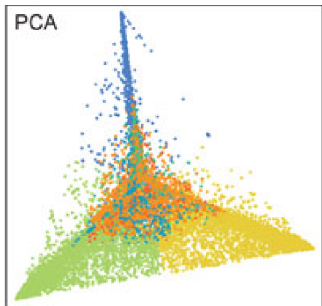
$$L_{TR}: d(q_n, q_m) + \alpha \leq d(q_n, q_l)$$

Type semantics

Relative geometry

Why VISAtlas Avoids PCA and t-SNE

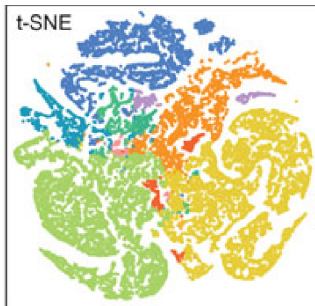
PCA



✓ Global structure

✗ Type semantics obscured

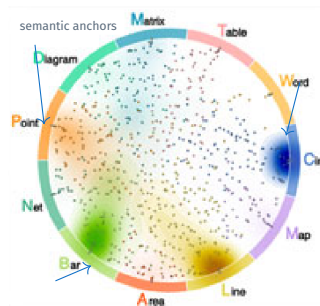
t-SNE



✓ Local clusters

✗ Inter-cluster distance unreliable

VISAtlas



$$P_n = \sum_i q_n^i v_i$$

✓ Fixed semantic meaning

PCA and t-SNE reveal structure; VISAtlas keeps the 11 visualization-type dimensions interpretable.

How VISAtlas Builds an Interactive Embedding Space

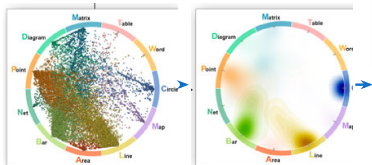
1 Projection

$$q_n = (q_n^1, \dots, q_n^{11})$$

$$P_n = \sum_i q_n^i v_i$$

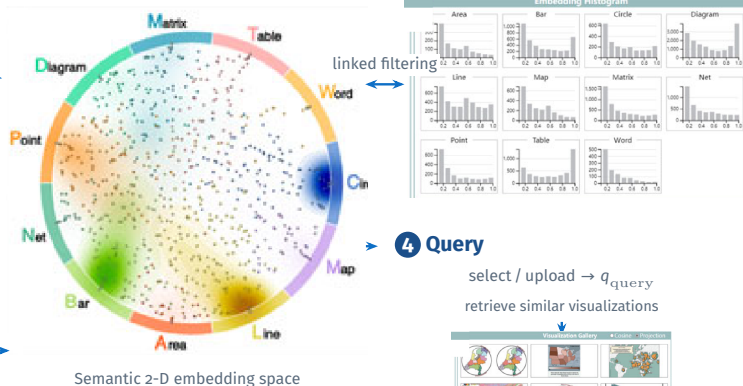
larger $q_n^i \rightarrow$ closer to anchor v_i

2 Rendering



overplotting $\rightarrow$ density + sampling

3 Inspection



4 Query

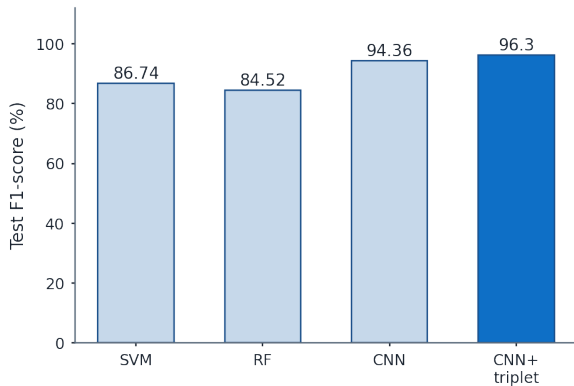
select / upload $\rightarrow q_{\text{query}}$
retrieve similar visualizations



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Evaluation: Case Studies

Does Real-World Data Actually Help?

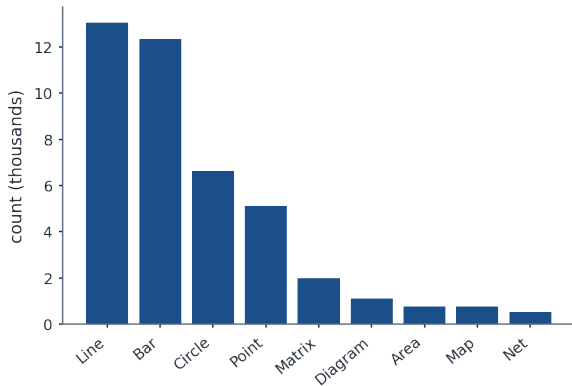


- Overall F1: **82.3%** (synthetic only) → **94.4%** (synthetic + real), same CNN
- Biggest gains: **Map** and **Diagram** — real maps have texture and noise a clean D3 render never has
- CNNs clearly beat classic ML (SVM, Random Forest)
- Triplet loss adds another ~2 points on top of plain classification

Case Study 1 — Comparing Collections

- **Beagle** (web-crawled, 41k): skews heavily to Line, Bar, Circle, Point
- **VIS3oK** (IEEE VIS papers, 29k): more balanced, but Diagram-heavy
- **Data2Vis** (synthetic, 4.3k): narrow — only Point, Line, Area, Bar

Counts we observed ourselves, running the Beagle dataset locally — confirms the paper's claim.



Study 2: finding composite visualizations

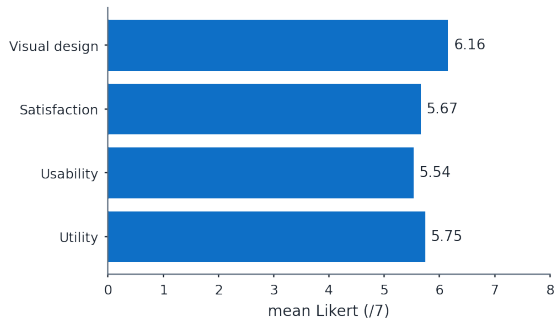
- Filter for points *without* a dominant embedding dimension (e.g. Matrix $\approx$ Table $\approx$ 0.4–0.5)
- Found: table visualizations with embedded heatmap cells — genuinely composite designs a keyword search would never surface

Study 3: querying with real vs. hand-drawn sketches

- Robust to different colormaps and layouts for *real* images
- A rough hand-drawn pie chart came back misclassified as a spiral line graph — wiggly strokes fall outside the training distribution

User Study: Does It Actually Help People?

- 12 postgraduate participants, 3 tasks mirroring the 3 case studies
- 7-point Likert scale; **every question averaged above 5/7**
- Best-rated: visual design of the Embedding Overview (mean 6.16)
- Weak spot: the Embedding Histogram's multi-step filtering took time to learn

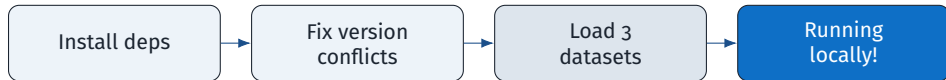


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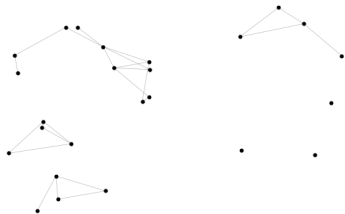
Our Reproduction & Discussion

We Didn't Just Read It — We Ran It

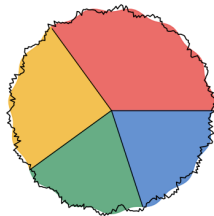
- Installed Python 3.9 + TensorFlow 2.8, Node/Vite, and all three datasets locally on Windows
- Fixed several rough edges (dependency conflicts, hard-coded model paths) — a reminder that research code $\neq$ production code
- End result: a fully working local instance, classifying and retrieving live



Confirming the Paper's Own Documented Weak Spots



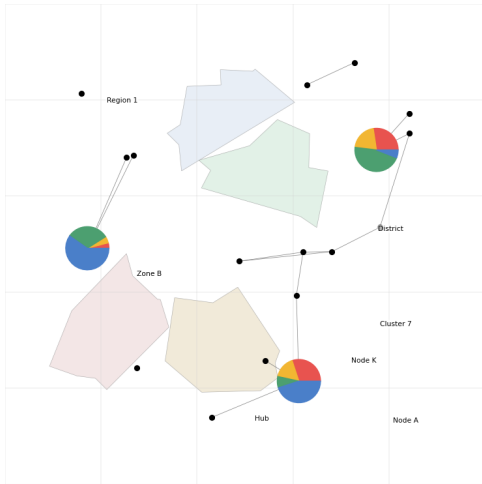
Thin-edge network graph



Simulated hand-drawn pie chart

Both images reproduced failure modes the paper itself describes: the sparse network was pulled toward *Point*, and the wobbly hand-drawn pie chart did not classify cleanly as *Circle*. Good — the paper's claims replicate.

Going Further: An Extreme 4-Way Composite

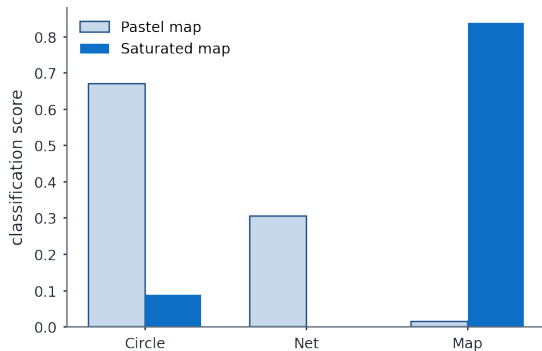
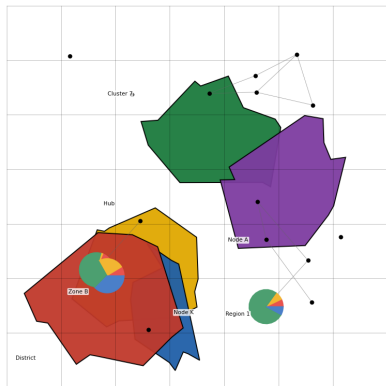


We mixed **four** visual vocabularies in one image: pastel map regions, a sparse network, three small pie charts, and scattered text labels.

Type	Score
Circle	0.671
Net	0.306
Map	0.015
Word	0.001

The paper only tested *two-way* composites (Matrix+Table) with comparably-weighted scores. Ours collapses hard onto one winner.

A Minimal-Pair Test: Same Content, Different Rendering



Map layer only: pastel → saturated & bold-bordered

Same network, same pie hubs, same text, same positions — **only** colour saturation and border weight on the map layer changed. The winning category flips completely.

What this suggests

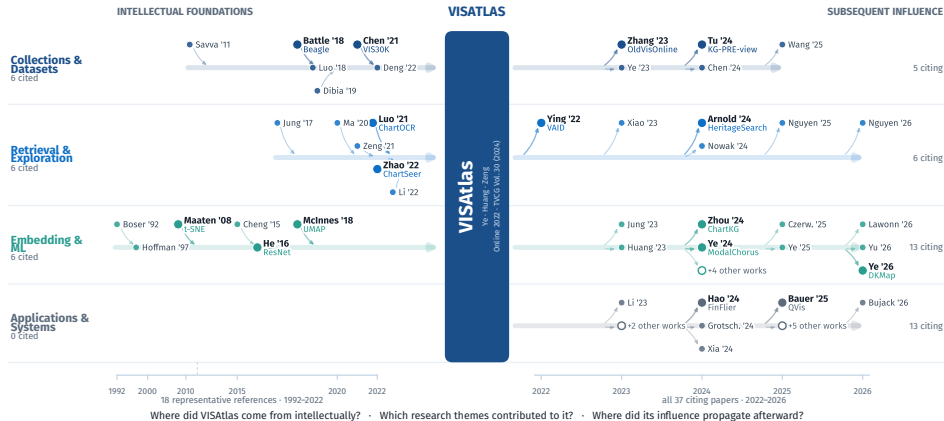
The embedding may respond more to **low-level visual salience** (colour saturation, contrast, edge density) than to the actual *structural semantics* of a chart type.

- The paper frames its embeddings as “human-interpretable” — our minimal-pair test suggests that interpretation can be steered by pure rendering style, independent of content
- This goes a step further than the paper’s own admission that the model doesn’t fully capture “colormaps and layout arrangements”
- **Discussion question:** if a CNN classifier can be flipped by style alone, how much should we trust embedding *distance* as a proxy for design *similarity*?

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Impact & Conclusion

From One Paper to a Research Programme



The core idea — contextual, anchor-based projection of neural embeddings — generalises from single-modality chart images (2022) to cross-modal LLM evaluation (2026), and to entirely new domains such as computational art criticism.

1. VISAtlas shows that **visual appearance**, not just metadata, is a legitimate and useful axis for exploring large chart collections
2. Its main trick — a **fixed, human-labelled anchor layout** (RadViz + CNN) — keeps a high-dimensional embedding interpretable, at the cost of distorting true distances
3. Its own authors, and our experiments, agree: the taxonomy and CNN backbone are still **too coarse** to fully capture design similarity
4. The idea proved durable: it is the direct ancestor of a still-active line of multimodal-embedding-visualization research

Thank you!

Questions?